33 Troubleshooting

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Topic Highlights

Logical/Analytical Troubleshooting Framework
The Seven-Step Troubleshooting Procedure
Vendor Assistance: Advantages and Pitfalls
Other Troubleshooting Methods

33.1 Introduction

Troubleshooting can be defined as the method used to determine why something is not working properly or is not providing an expected result. Troubleshooting methods can be applied to physical as well as non-physical problems. As with many practical skills, it is an art but also has an analytical or scientific basis. As such, basic troubleshooting is a trainable skill, while advanced troubleshooting is based on experience, developed skills, information, and a bit of art. While the discussion here centers on troubleshooting of instrumentation and control systems, the basic principles apply to broader classes of problems.

Troubleshooting normally begins with identifying that a problem exists and needs to be solved. The first steps typically involve applying a logical/analytical framework.

33.2 Logical/Analytical Troubleshooting Framework

A framework underlies a structure. Logical frameworks provide the basis for structured methods to troubleshoot problems. However, following a step-by-step method without first thinking through the problem is often ineffective. We also need to couple logical procedures with analytical thinking. To analyze information and determine how to proceed, we must combine logical deduction and induction with knowledge of the system, then sort through the information we have gathered regarding the problem. Often a logical/analytical framework does not produce the solution to a troubleshooting problem in just one pass. We usually have several iterations, which cause us to return to a previous step in the framework and go forward again. We can thus systematically eliminate possible solutions to our problem until we find the true solution.

33.2.1 Specific Troubleshooting Frameworks

Specific troubleshooting frameworks have been developed which apply to a particular instrument, class of instruments, system, or problem domain. For example, frameworks might be developed for a particular brand of analyzer, for all types of transmitters, for pressure control systems, or for grounding problems. When these match up with your system, you have a distinct starting point for troubleshooting.

Figure 33-1, for example, illustrates a specific problem domain troubleshooting flowchart, or tree, for transmitters.

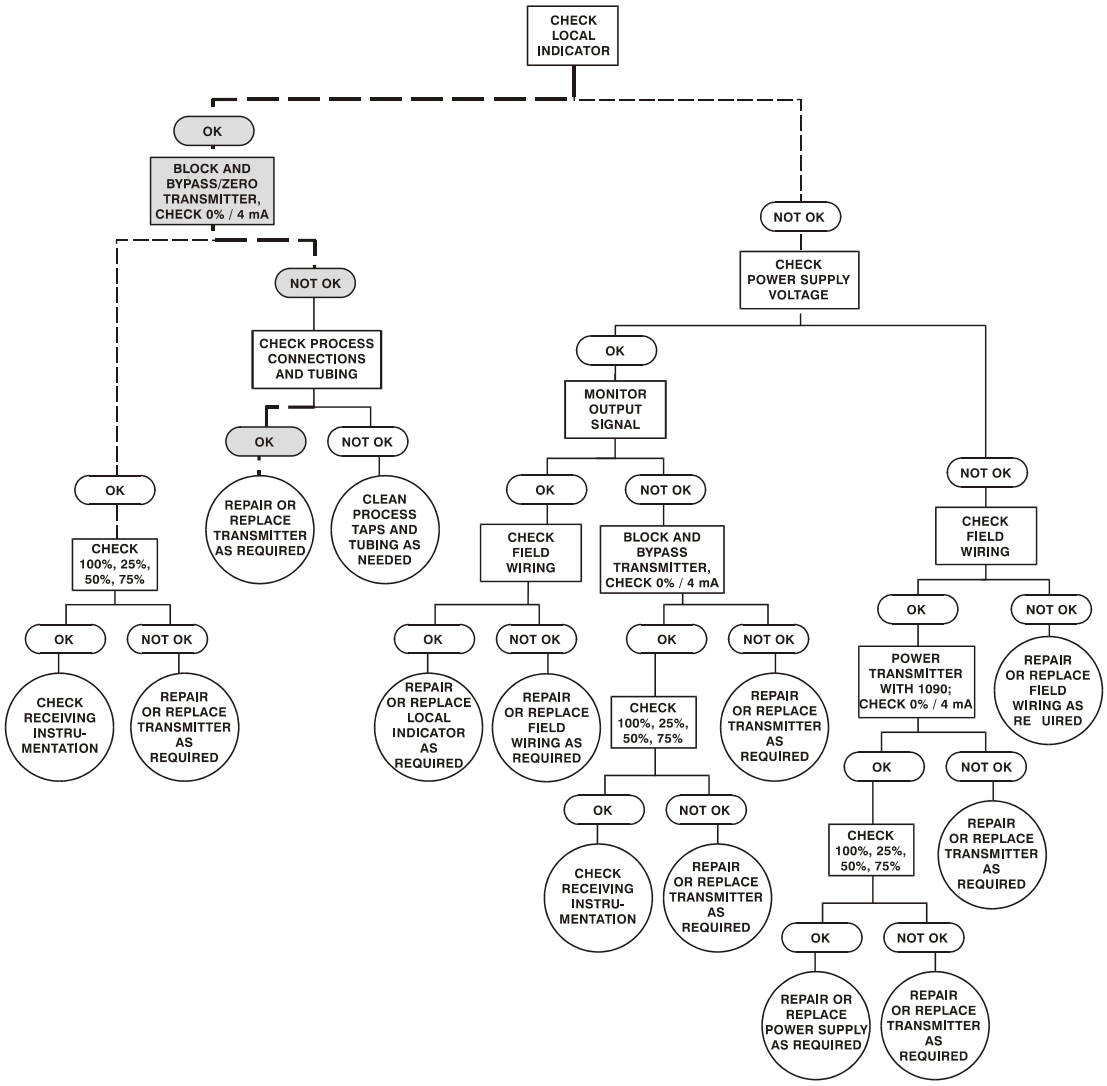


Figure 33-1: Specific Troubleshooting Framework Example

33.2.2 Generic Logical/Analytical Frameworks

Since we do not always have a specific structured framework available, we need a more general or generic framework that will apply to a broad class of problems. Figure 33-2 depicts this type of framework as a flowchart.

The framework shown in Figure 33-2, while efficient, does leave out some important safety-related tasks and company procedural requirements associated with or related to the troubleshooting process. A few important points should be made here:

- Troubleshooting increases the safety risk to the troubleshooter due to troubleshooting actions that involved on-line systems, energized systems, and moving parts.
- Always make sure what you are doing is safe for you, your fellow workers, and your facility.
- Follow company procedures.

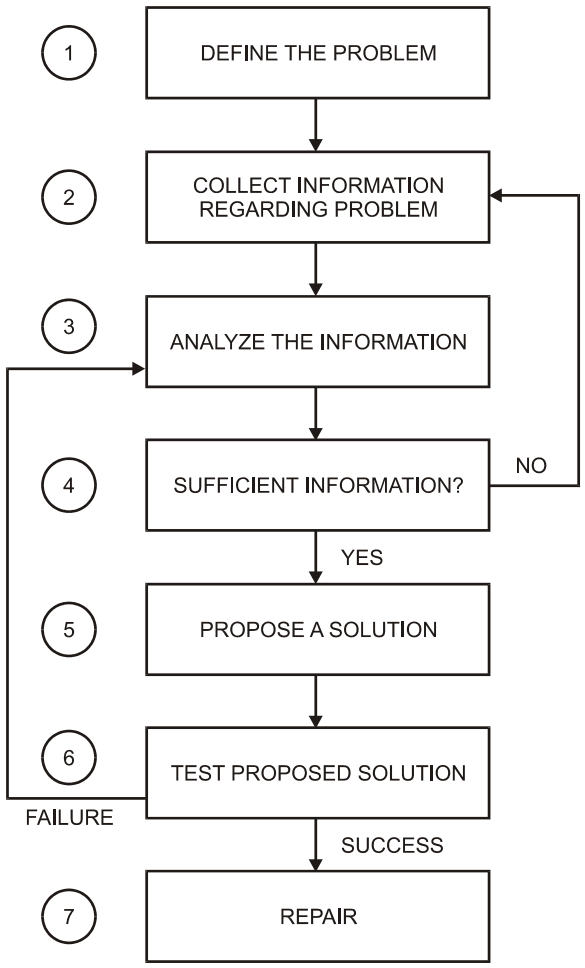


Figure 33-2: General Troubleshooting Framework Flowchart

- Get the proper permits, and follow their requirements.
- Always communicate your actions with the operator in charge and other involved people.
- Never put any part of your body anywhere that you do not know exactly what is there.
- Never take unnecessary risks. The life you save may be your own!

33.3 The Seven-Step Troubleshooting Procedure

The following seven-step procedure as illustrated in Figure 33-2 provides a generic, structured approach to troubleshooting.

33.3.1 Step 1: Define the Problem

You cannot solve a problem if you do not know what the problem is. The problem definition is the starting point. Get it wrong, and you will stray off of the path to the solution to the problem.

Communication

When defining the problem, listen carefully and allow the person(s) reporting the problem to you to provide a complete report of the problem as he or she sees it. The art of listening is a key element of troubleshooting. After listening carefully, ask clear and concise questions. All information has a sub-

jective aspect to it. When trying to identify a problem, you have to strip away subjective elements and get to the meat of the situation.

Avoid high-level technical terms or "technobabble." A troubleshooter must be able speak the "language" of the person reporting the problem. This means understanding the process; the plant physical layout; instrument locations; process functions as they are known in the plant; and the "dialect" of abbreviations, slang, and technical words commonly used in the plant. Some of this is generic to process plants in general, while some is specific to the plant in question.

33.3.2 Step 2: Collect Additional Information Regarding the Problem

Once a problem has been defined, you should then collect additional information. This step may overlap with Step 1, and, for simple problems, these two steps may even be the same. For complex or sophisticated problems, however, collecting information is a more distinct stage.

Develop a strategy or plan of action for collecting information. This plan should include determining where in the system you will begin to collect information, what sources will be used, and how the information will be organized. Information gathering typically moves from general to specific, though there may be several iterations of this. In other words, you are continually working to narrow down the problem domain.

Symptoms

The information you gather typically consists of symptoms (what is wrong with the system) as well as what is working properly. Primary symptoms are directly related to the cause of the problem at hand. Secondary symptoms are downstream effects—that is, not directly resulting from what is causing the problem. Differentiation between primary and secondary symptoms can be the key to localizing the cause of the problem.

Interviews and Collecting Information

Typically, a large part of your information gathering will be in the form of interviews with the person(s) who reported the problem and with any other people who may have relevant information. People skills are important here; good communication skills, the use of tact, and non-judgmental and objective approaches can be key to getting useful information.

Then, you review the instrument or system's performance from the control system's faceplates, trend recorders, summaries, operating logs, alarm logs, recorders, and system self-diagnostics. System drawings and documentation can also provide additional information.

Inspection

Next, you may inspect the instrument(s) that are suspected of being faulty, or other local instruments (such as pressure gauges, temperature gauges, sight glasses, and local indicators), to see if there are any other indications that might shed light on the matter.

History

Historical records can also provide useful information. Your facility's maintenance management system (MMS) may contain information regarding the failed system or ones like it. Also, check with others who have worked on the instrument or system.

Beyond the Obvious

If there are no obvious answers, testing may be in order. Plan your testing to ensure it is done safely and gets you the information you need with a minimum of intrusion. When you test by manipulating the system, plan to test or manipulate only one variable at a time. If you alter more than one, you might solve the problem, but be unable to identify what fixed the problem. Always make the minimum manipulation necessary to obtain the desired information. This minimizes the potential upset to the process.

33.3.3 STEP 3: Analyze the Information

Once you have collected information, you must start analyzing it to see if you have enough to propose a solution. Begin by organizing what you have collected.

You then can analyze the problem by reviewing what you already know, plus the new information you have gathered—connecting causes and effects, exploring causal chains, applying "If/Then" and "If/Then Not" logic, applying the process of elimination, and applying other relevant analytical or logical methods.

Case-Based Reasoning

Probably the first analytical technique that you will use is past experience—you have seen the same problem before. If you have seen this situation or case before, then you know a possible solution. Note that we say "a possible solution" because similar symptoms sometimes have different causes and hence different solutions.

"Similar To" Analysis

Compare the system you are working on to similar systems you have worked on in the past. For example, a pressure transmitter, a differential pressure transmitter, and a differential pressure level transmitter are similar instruments. Different PLC brands often have considerable similarities. RS-485 communication links are similar even on very different source and destination instruments. Similar instruments and systems operate on the same basic principles and have potentially similar problems and solutions.

"What, Where, When" Analysis

This type of analysis resembles the "Twenty Questions" game. You ask questions about what the gathered information may tell you. These are questions such as:

- What is working?
- What is not working?
- What is a cause of an effect (symptom) and what is not?
- What has changed?
- What has not changed?
- Where does the problem occur?
- Where does it not occur?
- When did the problem occur?
- When did it not occur?

Patterns

Symptoms can sometimes be complex and can be distributed over time. Looking for patterns either in symptom actions or lack of action or in time of occurrence can sometimes help in the analysis of symptoms.

Basic Principles

Apply basic scientific principles to analyze data—such as electrical current can only flow certain ways. Ohm's and Kirchhoff's Laws always work, mass and energy always balance, physical properties, etc.

The Manual
When in doubt, read the manual! It may have information on circuits, system analysis, or troubleshooting that can lead to a solution. It may also provide voltage, current, or indicator readings, test points, and analytical procedures. Often manuals have troubleshooting tables or charts to assist you.

Logical Methods
Now you will need a logical approach to make this iterative procedure successful. Several approaches, such the linear approach and the "Divide and Conquer" method, may apply.

First, try the linear or walk-through approach. This is a step-by-step process (illustrated in Figure 33-3) that you follow through a system. The first step is to decide on an entry point. If the entry point tests correctly, then you test the next point downstream in a linear signal path. If this test point is all right, then you choose the next point downstream of the previous test point, and so on. Conversely, if the entry point is found to be bad, choose the next entry point upstream and begin the process again. As you move downstream or upstream, each step narrows down the possibilities. Any branches must be tested at the first likely point downstream of the branch.

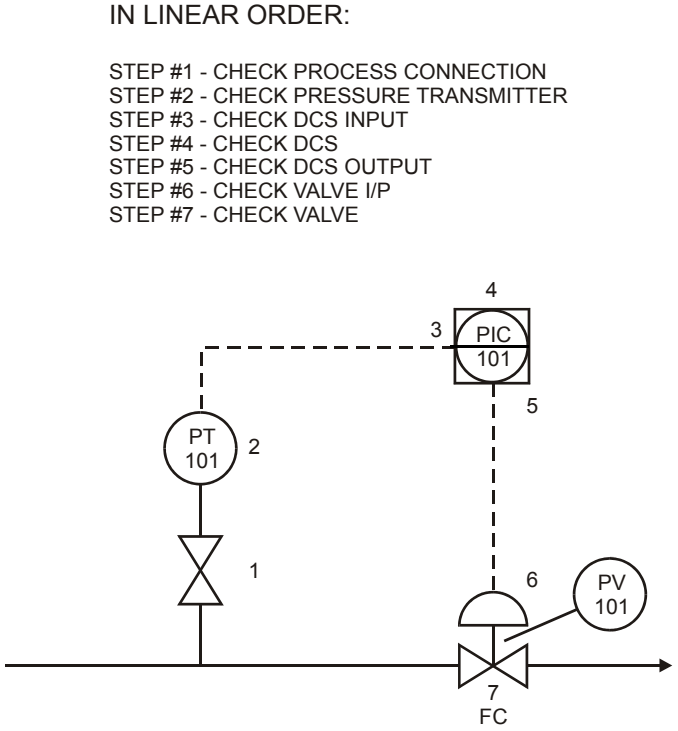


Figure 33-3: Linear Troubleshooting Approach

Second, "Divide and Conquer" is a general approach that is illustrated in Figure 33-4. You choose a likely point, or commonly the midpoint of the system, and test it. If it tests bad, then the upstream section of the system contains the faulty part. The upstream section is then divided in two parts and the system is tested at the dividing point. If, on the other hand, the test is good, the downstream section contains the bad part. Divided that in two, and so on, until the cause of the problem has been found.

33.3.4 Step 4: Determine Sufficiency of Information

When you are gathering information, how do you know that you have enough? Can you determine a cause and propose a solution to solve the problem? This is a decision point for moving on to the next step of proposing a solution.

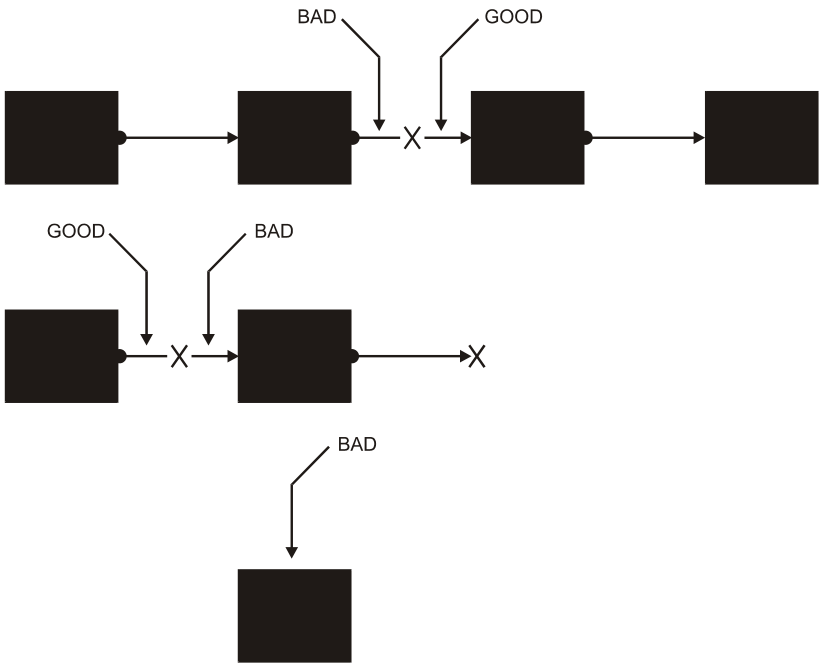


Figure 33-4: "Divide and Conquer" Troubleshooting Technique (Courtesy of Control Magazine)

33.3.5 Step 5: Propose a Solution

When you believe you have determined the cause of the problem, propose a solution. In fact, you may propose several solutions, based on your analysis. Usually the proposed solution will be to remove and replace (or repair) a bad part. In some cases, however, your proposal may not offer complete certainty of solving the problem and will have to be tested. If you have several possible solutions, propose them in the order of their probability of success. If this is roughly equal, or other operational limitations come into play, you can use other criteria. You might propose solutions in the order of the easiest to the most difficult. In other cases, there may be cost penalties (labor costs, cost of consumable parts, lost production, etc.) associated with trying various solutions; you may propose to try the least costly but viable option.

Do not try several solutions at once. This is called the "Shotgun Approach" and will typically only confuse the issue. Management will sometimes push for a shotgun approach due to time or operational constraints, but you should resist it. With a little analytical work, you may be able to solve the problem and meet management constraints at a lower cost. With the shotgun approach, you may find you do not know what fixed the problem, and it will be more costly both immediately and in the long term. If you do not know what fixed the problem, you may be doomed to repeat it.

Do not rush to a compromise solution proposed by a "committee," either. Consider the well-known "Trip to Abilene" story, illustrating the "group think" concept that is the opposite of synergy. In the story, some people are considering going to Abilene, though none of them really wants to go. They end up in Abilene, however, because everyone in the group figures everyone else wants to go to Abilene. This sometimes occurs in troubleshooting when a committee gets together to "assist" the troubleshooter and the committee gets side tracked by a trip to Abilene.

33.3.6 Step 6: Test The Proposed Solution

Once a solution, or combination of solutions, has been proposed, it must be tested to see if your analysis of the problem is correct.

Specific Versus General Solutions

However, be careful of specific solutions to more general problems. At this step, you must determine if the solution needed is more general than the specific one proposed. In most cases, a specific solution will be repairing or replacing the bad instrument, but that may not solve the problem.

But what if replacing an instrument only results in the new instrument going bad? Suppose an instrument with very long signal lines sustains damage from lightning transients. The specific solution would be replacing the instrument; the general solution might be to install transient protection on the instrument as well.

The Iterative Process

If the proposed and tested solution is not the correct one, then return to Step 3, "Analyze the Information." Where might you have gone astray? If you find your mistake, then move on to propose another solution. If not, move back to Step 2, "Collect Information." It is time to gather more information that will lead you to the real solution.

33.3.7 Step 7: The Repair

In the repair step, implement the solution you have proposed. In some cases, testing a solution results in the repair, as in replacing a transmitter, which both tests the solution and repairs the problem. Even in this case, there will generally be some additional work to be done, such as tagging, updating the database, and updating maintenance records, in order to complete the repair. Document the repair so future troubleshooting is made easier.

33.4 Vendor Assistance: Advantages and Pitfalls

Sometimes it is necessary to involve vendors in troubleshooting, either directly or by phone. Manufacturer's service personnel can be very helpful (and can provide a learning experience), but some are quick to blame other parts of the system (not their own) when they cannot find anything wrong—in some cases before they have even checked out their own system. Do not let vendors off the hook when trying to solve a problem just because they say it is not their equipment. Ask questions and make them justify their position.

33.5 Other Troubleshooting Methods

There are other types of troubleshooting methods available which complement the logical/analytical framework. Some of these are Substitution, Fault Insertion, "Remove and Conquer," "Circle the Wagons," "Complex-to-Simple," "Trapping," Consultation, Intuition, and "Out-of-the-Box" Thinking.

33.5.1 The Substitution Method

The substitution method substitutes a known good component for a suspected bad component. For modularized systems or those with easily replaceable components, substitution may reveal the component that is the cause of the problem. First, define the problem and gather information and analyze just as you do in the generic troubleshooting framework. Then, select a likely replacement candidate and substitute a known good component for it. By substituting components until the problem is found, the substitution method may find problems even where there is no likely candidate or only a vague area of suspicion. One potential problem with substitution is a higher-level cause can damage the replacement component as soon as you install it, or a transient (such as lightning) may have caused the failure and your fix may only be temporary. The use of this method can raise the overall maintenance cost due to extra module cost and the cost of inventory of replacement modules.

33.5.2 The Fault Insertion Method

Sometimes you can insert a fault instead of a known good signal or value and see how the system responds. For example, when a software communication interface is periodically locking up, you may

suspect that the interface is not responding to an I/O timeout properly. You can test this by inserting a fault—an I/O timeout.

33.5.3 The "Remove and Conquer" Method

For loosely coupled systems that have multiple independent devices, removing the devices one at a time may help you find certain types of problems. For example, if a communication link with 10 independent devices talking to a computer is not communicating properly, you might remove the boxes one at a time until the offending box is found. Once the problem device has been detected and repaired, the removed devices should be reinstalled one at a time to see if any other problems occur. The "Remove and Conquer" technique is particularly useful when a communication system has been put together incorrectly or exceeds system design specifications. For example, there might be too many boxes on a communication link, cables that are too long, cable mismatches, wrong cable installation, impedance mismatches, or too many repeaters. In these situations, sections of the communication system can be disconnected to see what happens.

A similar technique, "Add Back and Conquer," means removing all the boxes and adding them back one by one until you find the cause of the problem.

33.5.4 The "Circle the Wagons" Method

When you believe the cause of a problem is external to the device or system, try the "Circle the Wagons" technique. Draw an imaginary circle or boundary around the device or system; then see what interfaces (such as signals, power, grounding, environmental, and EMI) cross the circle. Then, isolate and test each boundary crossing. Often this is just a mental exercise that helps you think about external influences, which then leads to a solution. Figures 33-5 and 33-6 illustrate this concept.

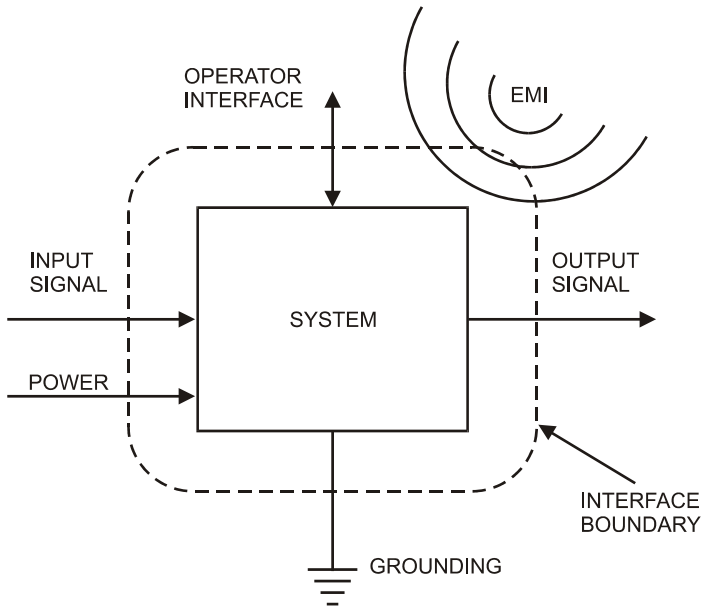


Figure 33-5: Circle the Wagons—Single Box Example

33.5.5 A Trapping We Shall Go

Sometimes an event that triggers or causes the problem is not alarmed, or is a transient, or happens so fast the system cannot catch it. This is somewhat like akin to having a mouse in your house. You generally cannot see it, but you can see what it has done.

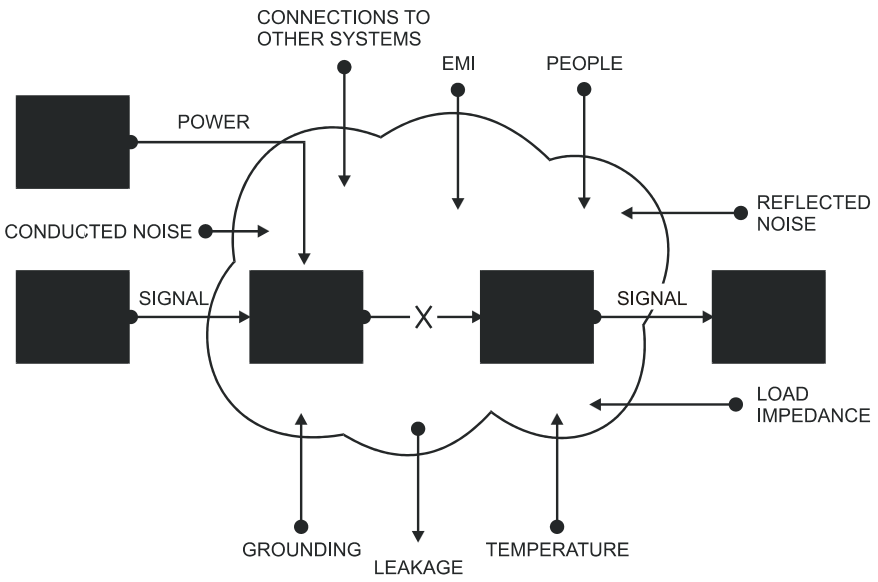


Figure 33-6: Circle the Wagons—Multiple Box Example (Courtesy of Control Magazine)

How do you catch the mouse? You set a trap. In sophisticated systems, you may have the ability to set additional alarms or identify trends to help track down the cause of the problem. For less sophisticated systems, you may have to use external test equipment or build a trap. If software is involved, you may have to build software traps that involve additional logic or code to detect the transient or bug.

33.5.6 The Complex-to-Simple Method

Many control loops and systems may have different levels of operation or complexity with varying levels of sophistication. One troubleshooting method is to break systems down from complex to simple. This involves finding the simple parts that function to make the whole. Once you find the simplest non-functioning "part," you can evaluate the non-functioning part or, if necessary, you can start at a simple known good part and "rebuild" the system until you can find the problem.

33.5.7 Consultation

Consultation, also known as the "third head" technique, means you use a third person, perhaps someone from another department, or an outside consultant, with advanced knowledge about the system or the principles for troubleshooting the problem. This person may not solve the problem but may ask questions that make the cause apparent or that spark fresh ideas for you. This process allows you to stand back during the discussions, which sometimes can help you distinguish trees from forest. The key is to know when you have reached the limitations of your investigation and need some additional help or insight.

33.5.8 Intuition

Intuition can be a powerful tool. What many people would call troubleshooting "intuition" certainly improves with experience. During troubleshooting or problem solving, threads of thought in your consciousness or sub-consciousness may develop, one of which may lead you to the solution. The more experience you have, the more threads develop during the troubleshooting process. Can you cultivate intuition? Experience suggests that you can, but success varies from person to person and from technique to technique. Find what works for you!

33.5.9 "Out-of-the-Box" Thinking

Difficult problems may require different approaches beyond normal or traditional troubleshooting methods. The term "out-of-the-box thinking" was a buzzword for organizational consultants during

the 1990s. "Out-of-the-box" thinking means approaching a problem from a new perspective, not being limited to the usual ways of thinking about it.

The problem in using this approach is that our troubleshooting "perspective" is generally developed along pragmatic lines, i.e., what has worked before; changing can sometimes be difficult.

How can you practice "out-of-the-box" thinking? How can you shift your perspective to find another way to solve the problem? Here are some questions that may help:

- Is there some other way to look at the problem?
- Can the problem be divided up in a different way?
- Can different principles be used to analyze the problem?
- Can analyzing what works rather than what does not work help to solve the problem?
- Can a different starting point be used to analyze the problem?
- Are you looking at too small a piece of the puzzle? Too big?
- Could any of the information on which the analysis is based be in error, misinterpreted, or looked at in a different way?
- Can the problem be conceptualized differently?
- Is there another "box" that that has similarities that might provide a different perspective?

33.6 Summary

While troubleshooting is an art, it is also based on scientific principles and is a developed skill that is both trainable and develops with quality experience. An organized, logical approach to troubleshooting is necessary to be successful and can be provided by following a logical framework, supplemented by generalized techniques such as Substitution, "Remove and Conquer," "Circle the Wagons," and "Out-of-the-Box" thinking.

33.7 References

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William L. "Bill" Mostia has 30+ years of experience in instrumentation, controls, safety, and electrical areas. He is currently a Senior Consultant with SIS-Tech Solutions, a leading consulting firm specializing in hazards & risk analysis, safety instrumented systems, and related engineering. He has worked for Amoco, Texas Eastman, Dow Chemical Co., exida, and as an independent consultant in instrument, electrical, and safety areas. He graduated from Texas A&M University with a BSEE. He is a professional engineer and a senior member of ISA. He is an active member of ISA and serves on a number of ISA standards committees including SP84, SP91, and various SP12 committees. He has published over 50 articles and papers and a book on troubleshooting, as well as been a contributor to several books on instrumentation.